Terraform-Task-2

1). Watch the **Terraform-02** video.
2). Execute all the **templates shown in the video**.

```
Last login: Wed Oct 28 08:36:11 on ttys000
rohithgoud@Rohiths-MacBook-Air ~ % terraform --version
Terraform v1.17.0
on darwin_arm64

Your version of Terraform is out of date! The latest version
is 1.17.1. You can update by downloading from https://developer.hashicorp.com/terraform/install

rohithgoud@Rohiths-MacBook-Air ~ % mkdir terraform-02
rohithgoud@Rohiths-MacBook-Air ~ % cd terraform-02
rohithgoud@Rohiths-MacBook-Air terraform-02 % touch main.tf
rohithgoud@Rohiths-MacBook-Air terraform-02 % open -e main.tf
rohithgoud@Rohiths-MacBook-Air terraform-02 % pwd
/Users/rohithgoud/terraform-02
rohithgoud@Rohiths-MacBook-Air terraform-02 % ls
main.tf
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform init
Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```

```
Default output format [json]: json
rohithgoud@Rohiths-MacBook-Air terraform-02 % aws sts get-caller-identity
{
    "UserId": "AIDATLUSGUBY4MGUQHONL",
    "Account": "231161110641",
    "Arn": "arn:aws:iam::231161110641:user/terraform-lab"
}
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform validate
Success! The configuration is valid.

rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform plan

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket.demo will be created
  + resource "aws_s3_bucket" "demo" {
      + acceleration_status         = (known after apply)
      + acl                         = (known after apply)
      + arn                         = (known after apply)
      + bucket                      = "terraform-02-demo-rohith-2026"
      + bucket_domain_name          = (known after apply)
      + bucket_prefix               = (known after apply)
      + bucket_regional_domain_name = (known after apply)
      + force_destroy               = false
      + hosted_zone_id              = (known after apply)
      + id                          = (known after apply)
      + object_lock_enabled         = (known after apply)
      + policy                      = (known after apply)
      + region                      = (known after apply)
      + request_payer               = (known after apply)
      + tags_all                    = (known after apply)
      + website_domain              = (known after apply)
      + website_endpoint            = (known after apply)

      + cors_rule (known after apply)

      + grant (known after apply)

      + lifecycle_rule (known after apply)

      + logging (known after apply)

      + object_lock_configuration (known after apply)

      + replication_configuration (known after apply)

      + server_side_encryption_configuration (known after apply)

      + versioning (known after apply)

      + website (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```

Terraform-Task-2

```
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket.demo will be created
  + resource "aws_s3_bucket" "demo" {
      + acceleration_status         = (known after apply)
      + acl                         = (known after apply)
      + arn                         = (known after apply)
      + bucket                      = "terraform-02-demo-rohith-2026"
      + bucket_domain_name          = (known after apply)
      + bucket_prefix               = (known after apply)
      + bucket_regional_domain_name = (known after apply)
      + force_destroy               = false
      + hosted_zone_id              = (known after apply)
      + id                          = (known after apply)
      + object_lock_enabled         = (known after apply)
      + policy                      = (known after apply)
      + region                      = (known after apply)
      + request_payer               = (known after apply)
      + tags_all                    = (known after apply)
      + website_domain              = (known after apply)
      + website_endpoint            = (known after apply)

      + cors_rule (known after apply)

      + grant (known after apply)

      + lifecycle_rule (known after apply)

      + logging (known after apply)

      + object_lock_configuration (known after apply)

      + replication_configuration (known after apply)

      + server_side_encryption_configuration (known after apply)

      + versioning (known after apply)

      + website (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_s3_bucket.demo: Creating...
aws_s3_bucket.demo: Creation complete after 1s [id=terraform-02-demo-rohith-2026]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
rohithgoud@Rohiths-MacBook-Air terraform-02 % aws s3 ls
2026-10-30 19:40:20 terraform-02-demo-rohith-2026
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform state list
aws_s3_bucket.demo
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```

```
rohithgoud@Rohiths-MacBook-Air terraform-02 % open -e main.tf
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform fmt
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform validate
Success! The configuration is valid.

rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform plan
aws_s3_bucket.demo: Refreshing state... [id=terraform-02-demo-rohith-2026]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_versioning.versioning will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = "terraform-02-demo-rohith-2026"
      + id     = (known after apply)

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.

─────────────────────────────────────────────────────────────────────────────

Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform apply
aws_s3_bucket.demo: Refreshing state... [id=terraform-02-demo-rohith-2026]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_versioning.versioning will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = "terraform-02-demo-rohith-2026"
      + id     = (known after apply)

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.
```

Terraform-Task-2

```
aws_s3_bucket.demo: Refreshing state... [id=terraform-02-demo-rohith-2026]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_versioning.versioning will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = "terraform-02-demo-rohith-2026"
      + id     = (known after apply)

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.

─────────────────────────────────────────────────────────────────────────────

Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform apply
aws_s3_bucket.demo: Refreshing state... [id=terraform-02-demo-rohith-2026]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_versioning.versioning will be created
  + resource "aws_s3_bucket_versioning" "versioning" {
      + bucket = "terraform-02-demo-rohith-2026"
      + id     = (known after apply)

      + versioning_configuration {
          + mfa_delete = (known after apply)
          + status     = "Enabled"
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_s3_bucket_versioning.versioning: Creating...
aws_s3_bucket_versioning.versioning: Creation complete after 2s [id=terraform-02-demo-rohith-2026]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
rohithgoud@Rohiths-MacBook-Air terraform-02 % aws s3api get-bucket-versioning \
  --bucket terraform-02-demo-rohith-2026 \
  --region ap-south-1
{
    "Status": "Enabled"
}
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```

```
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.

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Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
rohithgoud@Rohiths-MacBook-Air terraform-02 % aws s3api get-bucket-encryption \
--bucket terraform-02-demo-rohith-2026 \
--region ap-south-1
{
    "ServerSideEncryptionConfiguration": {
        "Rules": [
            {
                "ApplyServerSideEncryptionByDefault": {
                    "SSEAlgorithm": "AES256"
                },
                "BucketKeyEnabled": false,
                "BlockedEncryptionTypes": {
                    "EncryptionType": [
                        "SSE-C"
                    ]
                }
            }
        ]
    }
}
rohithgoud@Rohiths-MacBook-Air terraform-02 % terraform apply
aws_s3_bucket.demo: Refreshing state... [id=terraform-02-demo-rohith-2026]
aws_s3_bucket_versioning.versioning: Refreshing state... [id=terraform-02-demo-rohith-2026]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_server_side_encryption_configuration.encryption will be created
  + resource "aws_s3_bucket_server_side_encryption_configuration" "encryption" {
      + bucket = "terraform-02-demo-rohith-2026"
      + id     = (known after apply)

      + rule {
          + apply_server_side_encryption_by_default {
              + sse_algorithm     = "AES256"
                # (1 unchanged attribute hidden)
            }
        }
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_s3_bucket_server_side_encryption_configuration.encryption: Creating...
aws_s3_bucket_server_side_encryption_configuration.encryption: Creation complete after 1s [id=terraform-02-demo-rohith-2026]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```

```
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_public_access_block.public_access will be created
  + resource "aws_s3_bucket_public_access_block" "public_access" {
      + block_public_acls       = true
      + block_public_policy     = true
      + bucket                  = "terraform-02-demo-rohith-2026"
      + id                      = (known after apply)
      + ignore_public_acls      = true
      + restrict_public_buckets = true
    }

Plan: 1 to add, 0 to change, 0 to destroy.

─────────────────────────────────────────────────────────────────────────────

Note: You didn't use the -out option to save this plan, so Terraform can't
guarantee to take exactly these actions if you run "terraform apply" now.
aws_s3_bucket.demo: Refreshing state... [id=terraform-02-demo-rohith-2026]
aws_s3_bucket_versioning.versioning: Refreshing state... [id=terraform-02-demo-rohith-2026]
aws_s3_bucket_server_side_encryption_configuration.encryption: Refreshing state... [id=terraform-02-demo-rohith-2026]

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_s3_bucket_public_access_block.public_access will be created
  + resource "aws_s3_bucket_public_access_block" "public_access" {
      + block_public_acls       = true
      + block_public_policy     = true
      + bucket                  = "terraform-02-demo-rohith-2026"
      + id                      = (known after apply)
      + ignore_public_acls      = true
      + restrict_public_buckets = true
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_s3_bucket_public_access_block.public_access: Creating...
aws_s3_bucket_public_access_block.public_access: Creation complete after 0s [id=terraform-02-demo-rohith-2026]

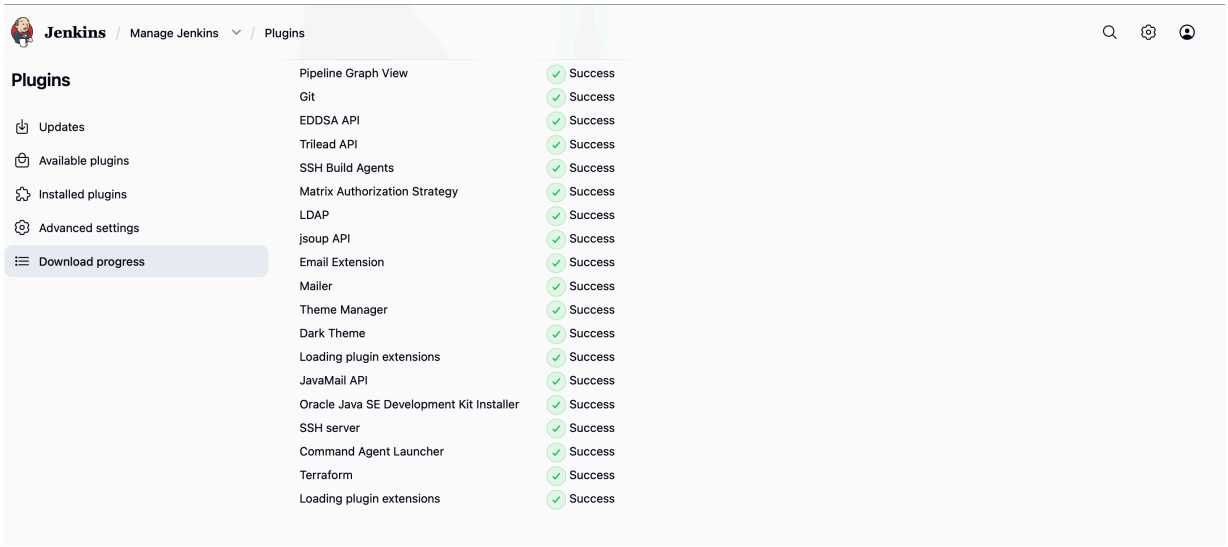
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
rohithgoud@Rohiths-MacBook-Air terraform-02 % aws s3api get-public-access-block \
--bucket terraform-02-demo-rohith-2026 \
--region ap-south-1
{
    "PublicAccessBlockConfiguration": {
        "BlockPublicAcls": true,
        "IgnorePublicAcls": true,
        "BlockPublicPolicy": true,
        "RestrictPublicBuckets": true
    }
}
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```

3). Integrate **Terraform in Jenkins** using the **Terraform plugin**.

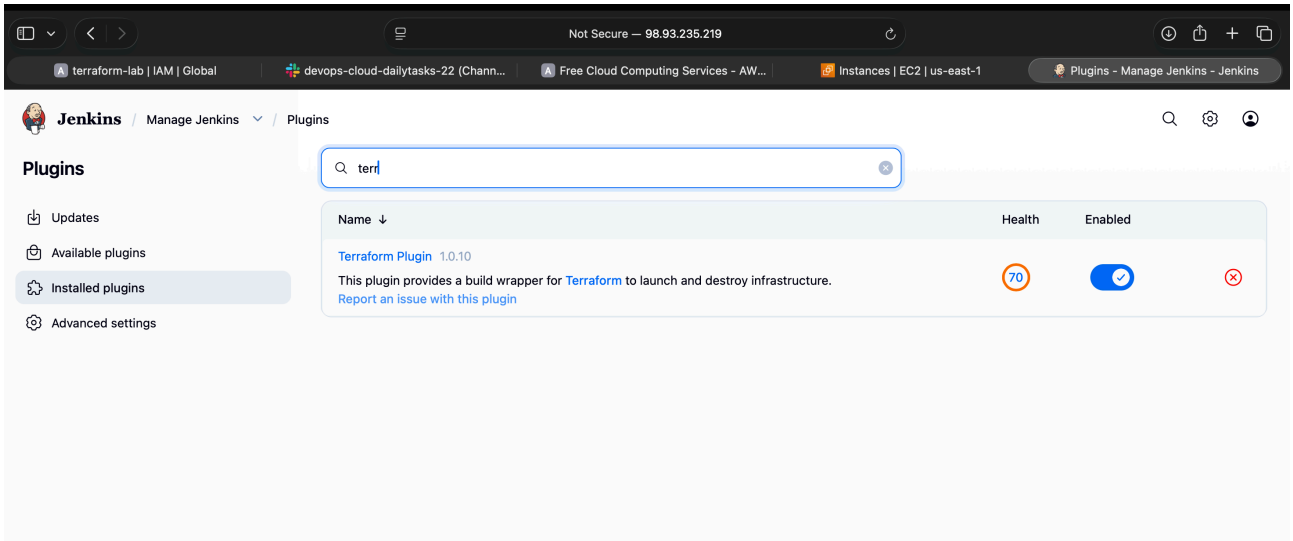
```
~
lines 1-21/21 (END)...skipping...
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-10-30 17:35:56 UTC; 18s ago
 Invocation: a354866bbaa14a3f86c1df4e16c067f1
   Main PID: 22811 (java)
      Tasks: 42 (limit: 627)
     Memory: 314.7M (peak: 325.6M)
        CPU: 21.142s
     CGroup: /system.slice/jenkins.service
             └─22811 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Oct 30 17:35:51 ip-10-0-26-238 jenkins[22811]: [LF]>
Oct 30 17:35:51 ip-10-0-26-238 jenkins[22811]: [LF]> ************************************************************
Oct 30 17:35:51 ip-10-0-26-238 jenkins[22811]: [LF]> ************************************************************
Oct 30 17:35:51 ip-10-0-26-238 jenkins[22811]: [LF]> ************************************************************
Oct 30 17:35:56 ip-10-0-26-238 jenkins[22811]: 2026-10-30 17:35:56.663+0000 [id=33]        INFO        jenkins.InitReactorRunner$1#onAttained: Completed initialization
Oct 30 17:35:56 ip-10-0-26-238 jenkins[22811]: 2026-10-30 17:35:56.693+0000 [id=24]        INFO        hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Oct 30 17:35:56 ip-10-0-26-238 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 30 17:35:56 ip-10-0-26-238 jenkins[22811]: 2026-10-30 17:35:56.939+0000 [id=50]        INFO        h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInsta>
Oct 30 17:35:56 ip-10-0-26-238 jenkins[22811]: 2026-10-30 17:35:56.941+0000 [id=50]        INFO        hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
Oct 30 17:36:02 ip-10-0-26-238 jenkins[22811]: 2026-10-30 17:36:02.073+0000 [id=69]        WARNING        h.n.DiskSpaceMonitorDescriptor#markNodeOfflineOrOnline: Making Built-In Node offline temporarily due>
~
```

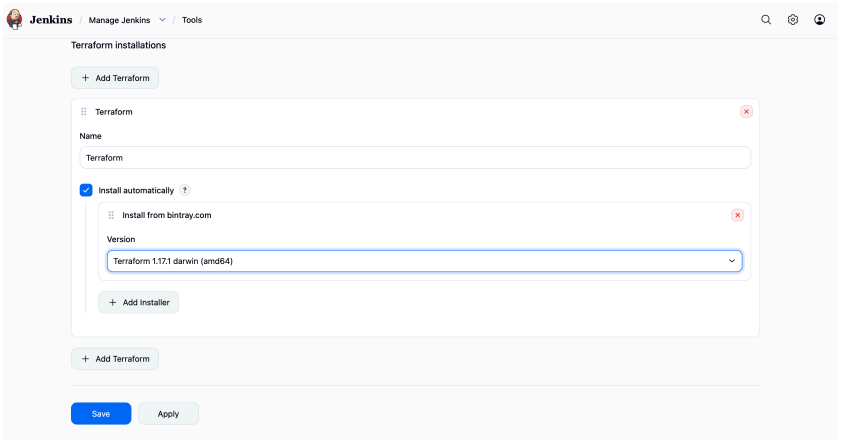
Terraform-Task-2

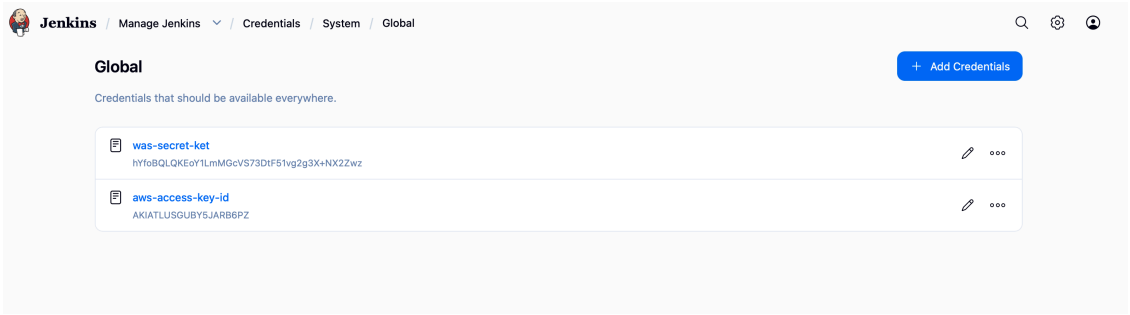


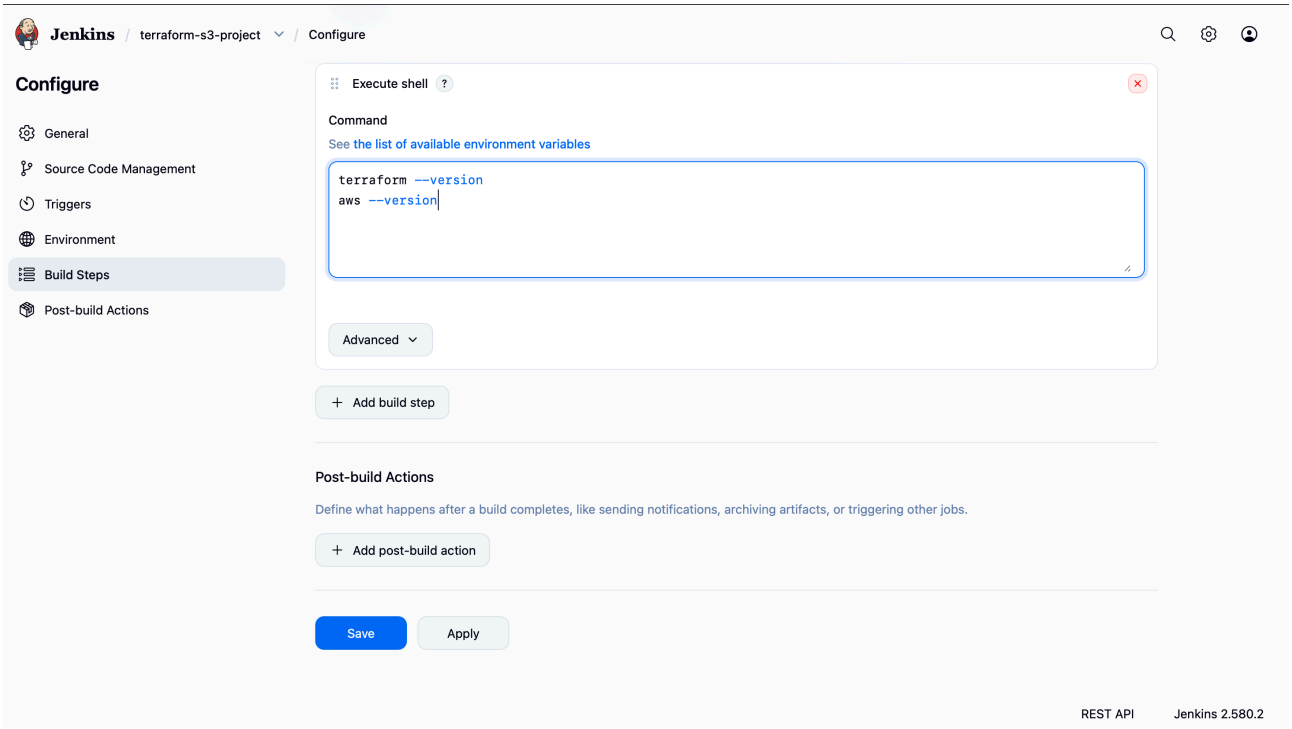




Terraform-Task-2







Terraform-Task-2




```
ubuntu@ip-10-0-26-238:~$ Read from remote host 98.93.235.219: Connection reset by peer
Connection to 98.93.235.219 closed.
client_loop: send disconnect: Broken pipe
rohithgoud@Rohiths-MacBook-Air terraform-02 % cd ~/terraform-02
ls -la
total 48
drwxr-xr-x   7 rohithgoud  staff   224 30 Oct 22:28 .
drwxr-x---+ 49 rohithgoud  staff  1568 30 Oct 22:28 ..
drwxr-xr-x   3 rohithgoud  staff    96 30 Oct 18:57 .terraform
-rw-r--r--   1 rohithgoud  staff  1407 30 Oct 18:57 .terraform.lock.hcl
-rw-r--r--@  1 rohithgoud  staff   708 30 Oct 22:28 main.tf
-rw-r--r--   1 rohithgoud  staff  5270 30 Oct 22:28 terraform.tfstate
-rw-r--r--   1 rohithgoud  staff  4505 30 Oct 22:28 terraform.tfstate.backup
rohithgoud@Rohiths-MacBook-Air terraform-02 %
```